27.

Elementary Statistics

Type - 1

1. The arithmetic mean of marks obtained by students of a class was 58. 20% of them achieved by the mean of the marks was 60 and the marks obtained by 30% the mean was 40. then find the mean of the marks obtained by remaining students?

(a) 65 (b) 66
(c) 68 (d) 70

RRB RPF Constable – 20/01/2019 (Shift-II)

Ans : (c) Let students number is the class =100
and mean of marks obtained by the remaining 50% students = x

According to the question-

$$20\times60+30\times40+50\times x=100\times58$$
$$1200+1200+50\times x=5800$$
$$2400+50x=5800$$
$$50x=5800-2400=3400$$
$$x=\frac{3400}{50}=68$$

Then mean of marks obtained by the remaining student is 68.

2. The mean of the observations $x, x+3, x+5, x+8, x+9$ is 9. What will be the mean of the last three observations.

(a) $\frac{32}{3}$ (b) $\frac{31}{3}$
(c) $\frac{35}{3}$ (d) $\frac{34}{3}$

RRB RPF Constable – 17/01/2019 (Shift-I)

Ans : (d) Mean $=\frac{\text{sum of total term}}{\text{number of terms}}$

$$9=\frac{x+x+3+x+5+x+8+x+9}{5}$$
$45 = 5x + 25$
$5x = 45 - 25$
$5x = 20$
$x = 4$

Valuve of last three terms
$x + 5 = 4 + 5 = 9$
$x + 8 = 12$
$x + 9 = 13$

$$\text{Mean}=\frac{9+12+13}{3}=\frac{34}{3}$$

3. Find the missing frequency 'x' in this figure. Given that the arithmetic mean is 28.

Profit %	0-10	10-20	20-30	30-40	40-50	50-60
Number of shops	12	18	27	x	17	6

(a) 20 (b) 15
(c) 12 (d) 24

RRB JE - 30/05/2019 (Shift-II)

Ans : (a)

Profit %	x	Shop's number (f)	f.x
0 – 10	5	12	60
10 – 20	15	18	270
20 – 30	25	27	675
30 – 40	35	x	35x
40 – 50	45	17	765
50 – 60	55	6	330
		$\sum f = 80+x$	$\sum fx = 2100+35x$

arithmetic mean $=\frac{\sum fx}{\sum f}$

$$28=\frac{2100+35x}{80+x}$$
$2240 + 28x = 2100 + 35x$
$7x = 140$
$x = 20$

4. If the mean of the given observation K, 2K + 1, 2K + 5, 2K + 9 is 30. then find the value of K?

(a) 15 (b) 5
(c) 12 (d) 20

RRB JE - 27/06/2019 (Shift-I)

Ans : (a) Mean $=\frac{\text{sum of total term}}{\text{number of terms}}$

$$30=\frac{K+2K+1+2K+5+2K+9}{4}$$
$120 = 7K + 15$
$7K = 105$
$K = 15$

5. **Since a mistake of 68 instead of 86 was typed, then the mean of the class was increased by 1/2. What is the total number of students?**

(a) 34 (b) 36
(c) 38 (d) 40

RRB RPF Constable – 19/01/2019 (Shift-II)

Ans : (b) Let number of students is x and mean of all class y mark

So total marks got by the students = xy

According to the question-

$xy + (86 - 68) = x\left(y + \frac{1}{2}\right)$

$\Rightarrow xy + 18 = xy + \frac{x}{2}$

$\Rightarrow \frac{x}{2} = 18$

$\Rightarrow x = 18 \times 2$

$\Rightarrow x = 36$

So students number = 36

6. **The mean of the five observation x, x +3, x + 4, x + 6 and x + 7 is 11. What will be the mean of the last three observations?**

(a) 12 (b) 12.67
(c) 19 (d) 13

RRB RPF SI – 16/01/2019 (Shift-II)

Ans : (b) Mean $= \frac{\text{sum of total term}}{\text{number of terms}} = \frac{\Sigma x}{N}$

$11 = \frac{x + (x+3) + (x+4) + (x+6) + (x+7)}{5}$

$55 = 5x + 20$

$35 = 5x$

$x = \frac{35}{5} = 7$

Mean of last three observation

$= \frac{(x+4) + (x+6) + (x+7)}{3}$

$= \frac{(7+4) + (7+6) + (7+7)}{3} = \frac{38}{3} = 12.67$

7. **The following list is given for the frequency distribution of diameter (D) of 101 steel balls.**

D (mm)	43	44	45	46	47	48
Number	13	15	22	21	16	14

Find the mean of the diameter in mm.

(a) 45.4 (b) 45.5
(c) 45.7 (d) 45.6

RRB Paramedical Exam –20/07/2019 (Shift-I)

Ans : (b) –

D (mm) (x)	43	44	45	46	47	48
n. (f)	13	15	22	21	16	14
fx	559	660	990	966	752	672

Mean $= \frac{\Sigma fx}{\Sigma f}$

$= \frac{559 + 660 + 990 + 966 + 752 + 672}{13 + 15 + 22 + 21 + 16 + 14}$

$= \frac{4599}{101} = 45.53$

8. **The mean of the marks obtained by 40 students in an examination is 72.5. Since a mistake of 48 instead of 84 was written. Find the right mean?**

(a) 71.3 (b) 72.4
(c) 77.5 (d) 73.4

RRB RPF SI – 13/01/2019 (Shift-III)

Ans : (d) As per question sum of marks obtained by 40 students right mean = 40 × 72.5 = 2900

According to the question-

right mean $= \frac{2900 + (84 - 48)}{40}$

$= \frac{2936}{40} = 73.4$

9. **If the mean of the given data 2, x, 7, 3, y, 9, 6, where x and y are constant is 6. If x and y replaced by 3x+1 and y+3 respectively, then mean increased by 2. Find the value of x?**

(a) 7 (b) 8
(c) 10 (d) 5

RRB Group-D – 20/09/2018 (Shift-III)

Ans : (d) Mean $= \frac{\text{sum of total marks}}{\text{number of marks}}$

$6 = \frac{2 + x + 7 + 3 + y + 9 + 6}{7}$

$x + y + 27 = 42$

$x + y = 42 - 27$

$x + y = 15$ .....(i)

According to the question,

$\frac{27 + 3x + 1 + y + 3}{7} = 8$

$31 + 3x + y = 56$

$3x + y = 56 - 31$

$3x + y = 25$ .....(ii)

Subtracting equation (i) from equation (ii)–

$3x + y = 25$

$x + y = 15$

$2x = 10$

$x = 5$

10. The mean of the 8 smallest numbers in a group is 17. While the mean of the all numbers is 20. If the smallest eight numbers leave, then the mean of the reamining number is 22. So find out the total numbers in the group?

(a) 20 (b) 22
(c) 18 (d) 19

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (a) : Let total numbers in group = x

According to the question-

$8\times17+(x-8)22=x\times20$

$136+22x-176=20x$

$-40+22x=20x$

$2x=40$

$x=20$

Total number in group = 20

11. The mean of 21 observations (all defferent) is 40. If the meadian increased by 21 them the value of observations increased, the mean of observations will be.

(a) 50 (b) 50.5
(c) 30 (d) 45

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (a) 21 Total sum of observation = 21 × 40 = 840

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

↓ +21

If the value of median is increased to 21, the value of the observations increases.

increase value of observation = (21 – 11) × 21 = 210

Mean of observation = $\frac{\text{Total mean increase value}}{21}$

$=\frac{840+210}{21}=50$

12. The mean of the 8 smallest numbers in a group is 12.5. While the mean of all 14 numbers in the group is 14. Find the mean of the 6 greatest numbers?

(a) 16.50 (b) 16.00
(c) 17.00 (d) 15.50

RRB Group-D – 10/10/2018 (Shift-I)

Ans : (b)

Mean of the 8 smallest number in the group = 12.5

sum of the smallest eight number = 8 × 12.5 = 100

mean of all 14 number = 14

14 sum of 14 number = 14 × 14 = 196

sum of 6 largest number = 196–100 = 96

mean of 6 largest number = $\frac{96}{6}=16$

So intended mean

13. The arithmetic mean of a set of numbers is 12. The mean of another set of numbers is 15. If combined mean of both sets is 12.5. Then find the ratio of frequency of both sets?

(a) 3 : 1 (b) 5 : 1
(c) 3 : 2 (d) 5 : 2

RRB Group-D – 24/10/2018 (Shift-II)

Ans. (b) : Let the number of terms in the first set of number (frequency) = n_1

and the number of terms in the second set of number (frequency)= n_2

According to the question–

$\Rightarrow 12\times n_1+15\times n_2=(n_1+n_2)\times12.5$

$\Rightarrow 12n_1+15n_2=12.5n_1+12.5n_2$

$\Rightarrow 15n_2-12.5n_2=12.5n_1-12n_1$

$\Rightarrow 2.5n_2=.5n_1\Rightarrow\frac{n_1}{n_2}=\frac{5}{1}$

So ratio both group's frequency = 5 : 1

14. Find the median of the given observation – 67, 34, 57, 32, 12, 92, 51, 62, 62 57, 93 and 5

(a) 56.5 (b) 32
(c) 57 (d) 62

RRB Group-D – 06/12/2018 (Shift-III)

Ans. (c) : Writing the given numbers in ascending order

5, 12, 32, 34, 51, $\boxed{57}$, $\boxed{57}$, 62, 62, 67, 92, 93

total numbers (n) =

$$\text{median}\left[\frac{\left(\frac{n}{2}\right)\text{st term}+\left(\frac{n}{2}\right)\text{st term}}{2}\right]$$

$$=\left[\frac{6^{th}\text{ term}+7^{th}\text{ term}}{2}\right]$$

$$=\left[\frac{57+57}{2}\right]=\frac{114}{2}=57$$

15. If the mean of the observation set $x_1, x_2 \ldots x_{10}$ is 20. Then find the mean of the set x_1+4, $x_2+8\ldots x_{10}+40$?

(a) 34 (b) 32
(c) 42 (d) 52

RRB Group-D – 15/11/2018 (Shift-I)

Ans : (c) $\frac{x_1+x_2+x_3+----x_{10}}{10}=20$

$x_1+x_2+x_3+----x_{10}=200$ ----(i)

then mean =

$$\frac{(x_1+4)+(x_2+8)+(x_3+12)+----(x_{10}+40)}{10}$$

$$=\frac{x_1+x_2+x_3+----x_{10}+4+8+12+---40}{10}$$

$$=\frac{200+5(4+40)}{10} \quad \text{[from equation(i)]}$$

$$=\frac{200+220}{10}=\frac{420}{10}=42$$

16. In an examination, the mean of scores of the 36 students was 72.50. On revising the points, it was found that a student mark was written 65 by mistake instead of 56. Then find the carry mean?

(a) 71.50 (b) 72.25
(c) 72 (d) 73

RRB Group-D – 12/11/2018 (Shift-III)

Ans : (b) Right mean $=\frac{36\times72.50+(56-65)}{36}$

$$=\frac{2610-9}{36}=\frac{2601}{36}=72.25$$

17. The mean score of a group of 17 members was 15, where as mean score of another group of n members was 12. If their combined mean was 13.7, then find the value of n?

(a) 12 (b) 13
(c) 14 (d) 11

RRB Group-D – 12/11/2018 (Shift-III)

Ans : (b) Total score of 17 members = 17 ×15 = 255
Total score of n members = 12n
According to the question,

$$\frac{255+12n}{17+n}=13.7$$

13.7n + 232.9 = 255 + 12n
13.7 n – 12n = 255 – 232.9
1.7n = 22.1
n = 13

18. The mean weight of six children is 17.5 kg. If the person weights of five children are 14, 19, 23, 21 and 13 kg respectively, find the weight of the sixth child.

(a) 17 kg (b) 15 kg
(c) 16 kg (d) 18 kg

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (b) : Suppose the weight of sixth child is x kg,

$$\therefore 17.5=\frac{14+19+23+21+13+x}{6}$$

105.0 = 90 + x
x = 15
So weight of sixth child = 15kg

19. On the basis of given below data. Find out the mean of the digits?

marks	0	1	2	3	4	8
Number of students	6	5	4	3	2	5

(a) 3.3 (b) 2.8
(c) 2.5 (d) 3.2

RRB Group-D – 23/10/2018 (Shift-III)

Ans : (b)

Mark (x)	0	1	2	3	4	8	
Number of students (f)	6	5	4	3	2	5	Σf=25
fx	0	5	8	9	8	40	Σfx=70

Mean $=\frac{\Sigma fx}{\Sigma f}$

$$=\frac{70}{25}=2.8$$

20. The four numbers a, b, c and d are such that their overall average is 23. The average of a and b is 19.5. then find out the average of c and d?

(a) 26.5 (b) 25.5
(c) 24.5 (d) 27.5

RRB Group-D – 01/10/2018 (Shift-I)

Ans. (a) :
The four numbers are a, b, c, d which average is 23
Total of all numbers = 4 × 23 = 92
Total of a and b = 2 × 19.5 = 39.0
Sum of c and d = sum of 4 number -sum of 2 number
= 92 – 39
Sum of c and d = 53
average of c and d = $\frac{53}{2}$
average of c and d = 26.5

21. If the mean of the numbers 27+x, 31+x, 89+x, 107+x and 156 + x is 82. Then find the mean of the numbers 130+x, 126+x, 68+x, 50+x and 1+x?

(a) 30 (b) 75
(c) 50 (d) 70

RRB Group-D – 23/09/2018 (Shift-II)

Ans : (b) mean $=\frac{\text{sum of terms}}{\text{number of terms}}$

$$\Rightarrow\frac{27+x+31+x+89+x+107+x+156+x}{5}=82$$

$\Rightarrow 410+5x=410$

$\Rightarrow 5x=410-410$

$\Rightarrow 5x=0$

$$x=\frac{0}{5}=0$$

again

$$\frac{130+x+126+x+68+x+50+x+1+x}{5}(\text{Putting } x=0)$$

$$=\frac{130+126+68+50+1}{5}=\frac{375}{5}=75$$

intended mean = 75

22. If the arithmetic mean of the series $x_1, x_2, x_3, \ldots\ldots\ldots x_n$ is 1. Then find the mean of the series $\frac{x_1}{k}, \frac{x_2}{k}, \frac{x_3}{k}, \ldots\ldots\ldots \frac{x_n}{k}$ (where k > 0)?

(a) $\frac{1}{k}$ (b) k

(c) $\frac{2}{k}$ (d) 2k

RRB Group-D – 19/09/2018 (Shift-I)

Ans : (a) $\text{mean} = \frac{\text{sum of terms}}{\text{number of terms}}$

$$1=\frac{x_1+x_2+x_3+\text{------}x_n}{n}\text{-----(I)}$$

So,

$$\text{mean}=\frac{\left(\frac{x_1}{k}+\frac{x_2}{k}+\text{----}\frac{x_n}{k}\right)}{n}$$

$$\text{mean}=\frac{1}{k}\frac{(x_1+x_2+x_3+\text{---}x_n)}{n}$$

$$\text{mean}=\frac{1}{k}$$ {∵ from equation (I)}

23. If the arithmetic mean of 10 numbers is 35 and each number is increased by 2, find the mean of the new set of numbers.

(a) 28 (b) 34

(c) 40 (d) 37

RRB NTPC 05.04.2016 Shift : 3

Ans :(d) ∵ Two be added to each number

∴ Two will also be added to the parallel mean.

So new range mean = 35 + 2 = 37

24. The mean of the marks scored by 12 students of a class is 67.4. If the mean of another class consisting of 15 students is 72.3, find the their combined mean?

(a) 70.12 (b) 69.85

(c) 71.23 (d) 68.94

RRB NTPC 03.04.2016 Shift : 2

Ans :(a) Total of arks obtained by 12 students

15 Total of marks obtained by =15×72.3 = 1084.5

$$\text{Joint mean of both classes}=\frac{808.8+1084.5}{12+15}$$

$$\text{total of marks obtained by 12 students}=\frac{1893.3}{27}=70.12$$

25. The mean of the four different observations is 17.5. When a new observation whose value is 20 is added to it, then what will be its new mean?

(a) 18 (b) 17.5

(c) 19 (d) 18.5

RRB NTPC 28.03.2016 Shift : 2

Ans :(a) Mean of four observation = 17.5

Sum of four observation = 17.5 × 4 = 70.0

Adding new observation = 70 + 20 = 90

$$\text{New mean}=\frac{90}{5}=18$$

26. The arithmetic mean of 20 observation is 15.5, later it is found that an observation written 42 by mistake instead of 24. Find out the correct mean?

(a) 14 (b) 14.4

(c) 14.6 (d) 15

RRB NTPC 28.03.2016 Shift : 3

Ans : (c) Total sum of 20 inspection but was mistakenly read 42 instead of 24.

∴ Correct sum = 310 + 24 – 42 = 292

$$\text{So correct mean}=\frac{292}{20}=14.6$$

27. The mean of 22 observations is 10. When two more observations is included then new mean becomes 11. What will be the mean of two new observations?

(a) 19 (b) 20

(c) 21 (d) 22

RRB NTPC 19.04.2016 Shift : 1

Ans : (d) Mean of two new observation

$$=\frac{24\times11-22\times10}{2}$$

$$=\frac{264-220}{2}=\frac{44}{2}=22$$

28. The mean of 8 observations is 10. Three more observations is added the new mean becomes 12. Find out the mean of three new observations?

(a) 16 (b) 18

(c) 17.33 (d) 15

RRB NTPC 16.04.2016 Shift : 2

Ans : (c) Total sum of 8 observation = 8×10 = 80

Sum of observation on inclusion three other observation = 11×12 = 132

∴ Sum of three new observation = 132 – 80 = 52

$$\text{So mean of three new observations}=\frac{52}{3}=17.33$$

29. Find out the mean of the given below data–
$1, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{1}{4}, 2, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}$

(a) $\frac{15}{18}$ (b) $\frac{13}{18}$
(c) $\frac{7}{9}$ (d) $\frac{8}{9}$

RRB NTPC 11.04.2016 Shift : 1

Ans : (b) Mean $= \frac{\text{Sum of numbers}}{\text{number of numbers}}$

$$= \frac{1+\frac{1}{2}+\frac{1}{2}+\frac{3}{4}+\frac{1}{4}+2+\frac{1}{2}+\frac{1}{4}+\frac{3}{4}}{9}$$

$$= \frac{\frac{4+2+2+3+1+8+2+1+3}{4}}{9} = \frac{26}{36} = \frac{13}{18}$$

30. If the mean of the given data 18, 16, 22, 13, ? is 16 then find the value of '?'.
(a) 9 (b) 11
(c) 10 (d) 12

RRB NTPC 06.04.2016 Shift : 1

Ans : (b)

$$\text{Mean} = \frac{18+16+22+13+?}{5}$$

$$16 = \frac{69+?}{5} \Rightarrow 69+? = 16\times 5$$

$$? = 80-69 = 11$$

31. The mean of 9 observations is 18. Four more observations is added, the new mean becomes 19. What will be the mean of four new observations?
(a) 21.25 (b) 20.25
(c) 19 (d) 22

RRB NTPC 22.04.2016 Shift : 3

Ans : (a) Sum of 9 observations = 18 × 9 = 162
Sum of 13 observation = 19 × 13 = 247
So mean of four new observations $= \frac{247-162}{4}$

$$= \frac{85}{4} = 21.25$$

32. Find out the mean of first 6 prime numbers?
(a) 14/3 (b) 3
(c) 41/6 (d) 13/2

RRB NTPC 29.03.2016 Shift : 1

Ans : (c) First 6 prime numbers $= 2,3,5,7,11,13$

$$\therefore \text{ mean } = \frac{2+3+5+7+11+13}{6} = \frac{41}{6}$$

Type - 1

33. Find out the median of the given observations– (a + 4), (a – 3.5), (a – 2.5), (a – 3), (a – 2), (a + 0.5), (a + 5) and (a – 0.5).
(a) a – 1.25 (b) a – 2.5
(c) a – 1.5 (d) a – 0.75

RRB JE - 27/05/2019 (Shift-II)

Ans : (a) Writing the number in ascending –
(a–3.5), (a–3), (a–2.5), (a–2), (a–0.5), (a+0.5), (a+4), (a+5)
order (n) = 8(sum)

$$\text{median} = \frac{\left(\frac{n}{2}\right)\text{th term} + \left(\frac{n}{2}+1\right)\text{st terms}}{2}$$

$$= \frac{4^{th}\text{ term} + 5^{th}\text{ term}}{2}$$

$$= \frac{a-2+a-0.5}{2}$$

$$= \frac{2a-2.5}{2}$$

$$= \boxed{a-1.25}$$

34. Find out the median of the given below data?

In ₹ daily income	10-14	15-19	20-24	25-29	30-34	35-39
No. of employee	5	10	15	20	10	5

(a) 20 (b) 12.33
(c) 26.4 (d) 25.13

RRB JE - 30/05/2019 (Shift-I)

Ans : (d)

daily income in (₹)	Number of staff	Cumulative frequency
9.5 – 14.5	5	5
14.5 – 19.5	10	15
19.5 – 24.5	15	30 =cf
24.5 – 29.5	20	50
29.5 – 34.5	10	60
34.5 – 39.5	5	65

n = 65

$$\therefore \frac{n}{2} = \frac{65}{2} = 32.5$$

it is included in the cumulative frequency 50
So median class interval will be (24.5 – 29.5)
minimum limit (L) = 24.5
square height (h) = 5
frequency (f) = 20

The cumulative frequency of the class before the median class. (cf) = 30

$$\therefore \text{ median } = L + \left(\frac{\frac{n}{2} - cf}{f}\right) \times h$$

$$= 24.5 + \left(\frac{32.5 - 30}{20}\right) \times 5$$

$$= 24.5 + \frac{2.5}{4}$$

$$= 24.5 + 0.625$$

$$= 25.125 \approx 25.13$$

35. Find out the median of the positive factore of 48.

(a) 16 (b) 12
(c) 8 (d) 7

RRB RPF SI – 12/01/2019 (Shift-III)

Ans : (d) all positive factors of 48 are–

$1, 2, 3, 4, \boxed{6}, \boxed{8}, 12, 16, 24, 48$

n = 10 (even)

$$\therefore \text{ median} = \frac{10^{th}}{2} \text{ terms value } + \left(\frac{10}{2} + 1\right) \text{st terms value}$$

$$\frac{5^{th} \text{ terms value} + 6^{th} \text{ terms value}}{2}$$

$$= \frac{6+8}{2} = \frac{14}{2} = 7$$

36. What will be the median of the given below data 1.9, 8.4, 3.6, 5.8

(a) 5.1 (b) 4.7
(c) 5.2 (d) 5.6

RRB Paramedical Exam –21/07/2019 (Shift-III)

Ans : (b) Ascending order – 1.9, 3.6, 5. 8, 8.4

n = 4 (even)

$$\text{Median} = \frac{1}{2}\left[\frac{n^{nd}}{2} \text{ term} + \left(\frac{n}{2} + 1\right)^{st} \text{ term}\right]$$

$$\text{Median } = \frac{1}{2}\left[2\text{nd term} + 3^{rd} \text{ term}\right]$$

$$= \frac{1}{2}[3.6 + 5.8] = \frac{1}{2} \times 9.4 = 4.7$$

37. Find out the median of the given below data-

$$1, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{1}{4}, 2, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}$$

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{1}{6}$ (d) $\frac{3}{4}$

RRB RPF Constable – 22/01/2019 (Shift-III)

Ans : (b) Putting in ascending order to the figure-

$$\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, 1, 2$$

n = 9 (Odd)

$$\text{Median} = \left(\frac{n+1}{2}\right) \text{st term } = \left(\frac{9+1}{2}\right) \text{rd term}$$

$$= 5^{th} \text{ term } = \frac{1}{2}$$

38. What will be the median of the given below data 5, 2, 2, 7, 3 and 8

(a) 3.5 (b) 4.5
(c) 4 (d) 3

RRB RPF Constable – 22/01/2019 (Shift-III)

Ans : (c) 5, 2, 2, 7, 3, 8

Writing in ascending order

2, 2, 3, 5, 7, 8

n = 6

$$\frac{\text{Median} = \frac{n^{th}}{2} \text{ terms value} + \left(\frac{n}{2} + 1\right)^{st} \text{ term value}}{2}$$

$$= \frac{3+5}{2}$$

$$= \frac{8}{2} = 4$$

39. If a school has 50 girls of class X the height *survey data (in cm)* is as follows:

Length (in cm)	No. of girls
less than 140 cm	4
less than 145cm	11
less than 150 cm	29
less than 155 cm	40
less than 160 cm	46
less than 165 cm	50

Find the median of their height.

(a) 144.03 cm (b) 148.89 cm
(c) 145.03 cm (d) 149.03 cm

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (b)

length (in cm)	frequency	Cumulative frequency
135–140	4	4
140–145	7	11
145–150	18=f	29
150–155	$11=f_b$	40
155–160	6	46
160–165	4	50

$\Sigma f = N = 50$

$$\text{Median} = L + \left(\frac{\frac{N}{2} - f_b}{f}\right) \times i \quad \text{where } \frac{N}{2} = \frac{50}{2} = 25$$

$$= 145 + \frac{(25-11)}{18} \times 5 \qquad L - \text{low limit}$$

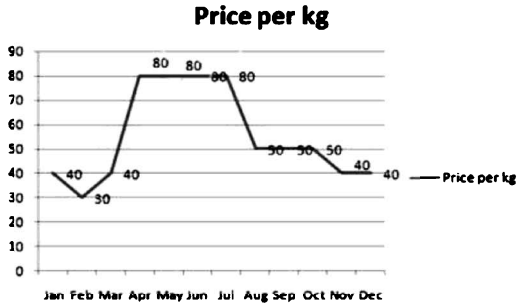
$$= 145 + \frac{70}{18} \qquad f - \text{original frequency}$$

$145 + 3.88 = 148.88 \approx 148.89$

i – original frequency

f_b – below frequency of original frequency

40. The following graph shows the price of 12 months of cabbage what is the median of their values?



(a) 55 (b) 40
(c) 50 (d) 60

RRB Group-D – 17/09/2018 (Shift-III)

Ans. (c) : Writing in ascending order to the given numbers, 30, 40, 40, 40, 40, 50, 50, 50, 80, 80, 80, 80

Total number (n) = 12(even)

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{th} \text{terms value} + \left(\frac{n}{2}+1\right)^{th} \text{terms value}}{2}$$

$$= \frac{\left(\frac{12}{2}\right)^{st} \text{terms} + \left(\frac{12}{2}+1\right)^{st} \text{terms}}{2}$$

$$= \frac{6^{th} \text{ term} + 7^{th} \text{ term}}{2} = \frac{50+50}{2} = 50$$

41. The mean of the given data 12, 13, 15, 18, X, 28, 18, 12, 6 and 8 is 15. Find out the median of the given data?

(a) 14.5 (b) 13.5
(c) 14 (d) 13

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (c) : mean of given 12, 13, 15, 18, x, 28, 18, 12, 6, 8

mean = 15

$$\frac{6+8+12+12+13+15+18+18+28+x}{10} = 15$$

$= 130 + x = 150$

$x = 150 - 130 = 20$

writing in ascending order to the figure

$\therefore 6, 8, 12, 12, \boxed{13, 15}, 18, 18, 20, 28$

terms number here = 10(even)

$$\text{mediam} \cdot \frac{1}{2}\left[\frac{n}{2}\text{th term} + \left(\frac{n}{2}+1\right)\text{th term}\right]$$

$$= \frac{1}{2}\left[\left(\frac{10}{2}\right)\text{th term} + \left(\frac{10}{2}+1\right)\text{th term}\right]$$

$$= \frac{1}{2}[5^{th} \text{ term} + 6^{th} \text{ term}]$$

$$\text{median} = \frac{13+15}{2} = \frac{28}{2} = 14$$

42. Find the median of the firt 10 numbers of the Fibonacci series: A Fibonacci number of that series is the sum of two last numbers first two Fibonacci numbers arte 0 and 1 respectively.

(a) 4 (b) 3
(c) 5 (d) 4.5

RRB Group-D – 11/10/2018 (Shift-III)

Ans : (a)

Number = 0, 1, 1, 2, 3, 5, 8, 13, 21, 34

terms number(n) = 10 (even)

$$\text{median} = \frac{(5\text{th term} + 6\text{th term})}{2} = \frac{3+5}{2} = 4$$

43. What will be the median of the given data– 5, 17, 68, 17, 32, 45, 64, 37, 93, 45, 78, 32, 35 and 45

(a) 41 (b) 44
(c) 42 (d) 43

RRB Group-D – 12/11/2018 (Shift-I)

Ans. (a) : Writing is ascending order of numbers-

5, 17, 17, 32, 32, 35, 37, 45, 45, 45, 64, 68, 78, 93

terms number = 14 (even)

$$\text{median} = \frac{1}{2}\left[\frac{n}{2}\text{th term} + \left(\frac{n}{2}+1\right)\text{th term}\right]$$

$$= \frac{1}{2}(7^{th} \text{ term} + 8^{th} \text{ term})$$

$$= \frac{1}{2}[37+45]$$

$$= \frac{82}{2} = 41$$

44. Find the median of the all prime number from 1 to 55.

(a) 22 (b) 20
(c) 21 (d) 19

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (c) According to the questions

All prime number of till 1 to 55 = 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53

n = 16 (even)

$$\therefore \text{ median} = \frac{\left(\frac{n}{2}\right)\text{th term} + \left(\frac{n}{2}+1\right)\text{th term}}{2}$$

$$= \frac{\left(\frac{16}{2}\right)\text{th term} + \left(\frac{16}{2}+1\right)\text{th term}}{2}$$

$$= \frac{8\text{th term} + 9\text{th term}}{2}$$

$$= \frac{19+23}{2}$$

$$= \frac{42}{2}$$

$$= 21$$

So median of all prime numbers between 1 to 55 = 21

45. Points scored by 12 people are 6, 17, 8, 9, 16, 10, 15, 21, 9, 11, 12 and 16 find the median.

(a) 11.5 (b) 11.6
(c) 10.4 (d) 12

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (a) : Putting in ascending order to the series.

6, 8, 9, 9, 10, 11, 12, 15, 16, 16, 17, 21

$\because$ n = 12 (even)

$$\text{Median} = \left[\frac{\left(\frac{n}{2}\right)\text{th term} + \left(\frac{n}{2}+1\right)\text{th term}}{2}\right]$$

$$= \left[\frac{\left(\frac{12}{2}\right)\text{th term} + \left(\frac{12}{2}+1\right)\text{th term}}{2}\right]$$

$$= \frac{6\text{th term} + 7\text{th term}}{2} = \frac{11+12}{2}$$

$$= \frac{23}{2} = 11.5$$

46. By a Kabbaddi team, in a series of matches obtained marks are given as follows–
17, 2, 7, 27, 15, 5, 14, 8, 10, 24, 48, 10, 8, 7, 18, 28 find the median of the marks obtained by the team?

(a) 11 (b) 12
(c) 16 (d) 15

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (b) : Writing in increasing order of digits– 2, 5, 7, 7, 8, 8, 10, 10, 14, 15, 17, 18, 24, 27, 28, 48

total terms number(n) is = 16(even)

$$\text{median} = \left[\left(\frac{n}{2}\right)^{\text{th}}\text{term} + \left(\frac{n}{2}+1\right)^{\text{th}}\text{term}\right] \div 2$$

$$= \frac{8\text{th term} + 9\text{th term}}{2} = \frac{10+14}{2} = \frac{24}{2} = 12$$

47. What will be the median of the given following data–
7, 21, 2, 17, 3, 13, 7, 4, 9, 7, 9

(a) 4 (b) 17
(c) 7 (d) 9

RRB NTPC 17.01.2017 Shift-3

Ans : (c) Ascending order to given figure–

2, 3, 4, 7, 7, $\boxed{7}$, 9, 9, 13, 17, 21

n = 11 (odd)

$$\text{median} = \frac{11+1}{2}\text{ st term} = 6 \text{ term} = 7$$

48. Find the median of the given following data–
9, 0, 2, 8, 5, 3, 5, 4, 1, 5, 2, 7

(a) 5 (b) 6.5
(c) 4.5 (d) 4

RRB NTPC 04.04.2016 Shift : 2

Ans : (c) Ascending order to the the given figure—

0, 1, 2, 2, 3, 4, 5, 5, 5, 7, 8, 9

n =12 even

$$\text{median} = \frac{1}{2}\left[\frac{12}{2}\text{th term} + \left(\frac{12}{2}+1\right)\text{st term}\right]$$

$$= \frac{1}{2}[6\text{th term} + 7\text{th term}]$$

$$= \frac{1}{2}(4+5) = \frac{9}{2} = 4.5$$

49. The median of the following numbers arranged in ascending order is 2.5, so find x?
0, 0, 1, 1, 2, 2, x, 3, 3, 4, 5, 7

(a) 2 (b) 3
(c) 4 (d) 0

RRB NTPC 03.04.2016 Shift : 2

Ans : (b) 0, 0, 1, 1, 2, 2, x, 3, 3, 4, 5, 7

n =12 (even)

$$\text{median} = \frac{1}{2}\left[\frac{n}{2}\text{th term} + \left(\frac{n}{2}+1\right)\text{th term}\right]$$

$$2.5 = \frac{1}{2}\left[\frac{12}{2}\text{th term} + \left(\frac{12}{2}+1\right)\text{th term}\right]$$

$$= \frac{1}{2}[6\text{th term} + 7\text{th term}]$$

$$2.5 = \frac{1}{2}[2+x]$$

$$\Rightarrow 2 + x = 5$$

$$\Rightarrow x = 3$$

50. For the given following data what will be the median? 25, 23, 26, 29, 31, 39 and 11

(a) 25 (b) 26
(c) 29 (d) 31

RRB NTPC 02.04.2016 Shift : 1

Ans : (b) Writing in ascending order,
11, 23, 25, 26, 29, 31, 39
Number of terms = 7 (odd)

$\therefore$ Median $= \left(\frac{7+1}{2}\right)$st tern = 4th term = 26

51. Find the median of the given number– 55, 53, 59, 56, 61, 69, and 31

(a) 55 (b) 56
(c) 59 (d) 61

RRB NTPC 02.04.2016 Shift : 2

Ans : (b) Writing in ascending order to the given numbers –

31, 53, 55, 56, 59, 61, 69
n = 7 (odd)

$$\text{median} = \frac{n+1}{2}\text{ st term} = \left(\frac{7+1}{2}\right)\text{st term}$$

= 4th term
= 56

52. What will be the median of the given number. 2, 3, 4, 3, 0, 5, 1, 1, 3, 2

(a) 0 (b) 3
(c) 2.5 (d) 2.4

RRB NTPC 31.03.2016 Shift : 3

Ans : (c) 2, 3, 4, 3, 0, 5, 1, 1, 3, 2 writing in ascending order – 0, 1, 1, 2, 2, 3, 3, 4, 5
n = 10 (even)

$$\text{median} = \frac{1}{2}\left[\frac{n}{2}\text{th term} + \left(\frac{n}{2}+1\right)\text{st term}\right]$$

$$= \frac{1}{2}\left[\frac{10}{2}\text{th term} + \left(\frac{10}{2}+1\right)\text{st term}\right]$$

$$= \frac{1}{2}[\text{5th term + 6th term}]$$

$$= \frac{1}{2}(3+2) = \frac{5}{2} = 2.5$$

53. If the numbers 3, 6, 7, 11, x, 15, 19, 20, 25 and 28 are in ascending order and median of these numbers are 13, then find the value of x?

(a) 11 (b) 12
(c) 13 (d) 14

RRB NTPC 28.03.2016 Shift : 3

Ans : (a)
Given number 3, 6, 7, 11, x, 15, 19, 20, 25, 28 here terms number = 10 (even)

$$\therefore \quad \text{Median} = \frac{\frac{n}{2}\text{th term} + \left(\frac{10}{2}+1\right)\text{th term}}{2}$$

$$13 = \frac{\left(\frac{10}{2}\right)\text{th term} + \left(\frac{10}{2}+1\right)\text{th term}}{2}$$

$$13 = \frac{\text{5th term + 6th term}}{2}$$

26 = x + 15
x = 11

54. What will be the median of the given data as follows -3, 4, 0, 4, -2, -5, 1, 7, 10, 5

(a) 2 (b) 2.5
(c) 2.75 (d) 3

RRB NTPC 11.04.2016 Shift : 2

Ans : (b) Writing in ascending order to the figure -
–5, –3, –2, 0, 1, 4, 4, 5, 7, 10
n = 10 (even)

$$\therefore \text{ median} = \frac{1}{2}\left[\frac{n}{2}\text{th term} + \left(\frac{n}{2}+1\right)\text{th term}\right]$$

$$= \frac{1}{2}[\text{5th term + 6th term}]$$

$$= \frac{1}{2}\times[1+4]$$

= 2.5

55. Find out the median of the given numbers as follows 3, 3, 5, 7, 8, 8, 8, 9, 11, 12, 12

(a) 9 (b) 7
(c) 8 (d) 12

RRB NTPC 19.01.2017 Shift : 3

Ans : (c) Writing in ascending order to the figure-
3, 3, 5, 7, 8, 8, 8, 9, 11, 12, 12
$\Rightarrow$ terms number(n) = 11
if terms number are odd then-

$$\text{median} = \frac{n+1}{2}\text{ th term}$$

$$= \frac{11+1}{2}\text{ th term} = \text{6th term} = 8$$

56. What will be the median of the given data- 87, 21, 53, 12, 86, 98, 23, 64, 87, 23, 23, 87, 56, 12, 53

(a) 53.5 (b) 54
(c) 53 (d) 56.5

RRB NTPC 28.04.2016 Shift : 3

Ans : (c) Arranging in ascending order to the given data–
12, 12, 21, 23, 23, 23, 53, 53, 56, 64, 86, 87, 87, 87, 98
Total number of digits (n) = 15 (odd)

$$\text{median} = \left(\frac{n+1}{2}\right)\text{st terms value}$$

$$\text{So, median} = \left(\frac{15+1}{2}\right)\text{st tems value} = \text{8th terms value}$$

So median = 53

57. The median of the following term was determined: 32, 12, 23, 17, 28, 25, 43. Later it was found that 17 was written by mistake instead of 29. Now what will be the changeable median?

(a) 29 (b) 17
(c) 23 (d) 28

RRB NTPC 29.04.2016 Shift : 1

Ans : (d) 32, 12, 23, 17, 28, 25, 43
arranging in ascending
12, 23, 25, 28, 29, 32, 43
n = 7 (odd)

$$\text{median} = \left(\frac{n+1}{2}\right)\text{th term}$$

$$= \left(\frac{7+1}{2}\right)\text{st term} = 4\text{th term} = 28$$

Type - 3

58. What will be the mode of the given data as follows– 3, 12, 4, 6, 8, 5, 4

(a) 5 (b) 8
(c) 4 (d) 3

RRB RPF SI – 06/01/2019 (Shift-II)

Ans : (c) Mode = the highest frequency number
Mode = 4

59. Find out the mode of the given distribution–

Class interval	1–10	10–20	20–30	30–40	40–50	50–60
Frequency	3	16	26	31	16	8

(a) 34.5 (b) 35
(c) 42 (d) 32.5

RRB JE - 27/06/2019 (Shift-I)

Ans : (d)

Class interval	1–10	10–20	20–30	30–40	40–50	50–60
frequency	3	16	(26) F_0	(31) F_1	(16) F_2	8

Most frequency class is (30-40)
So mode category(30-40)

$\therefore$ $L_1 = 30, L_2 = 40$
$F_1 = 31, F_2 = 16, F_0 = 26$

formula– $\text{Mode} = L_1 + \frac{(L_2 - L_1)(F_1 - F_0)}{2F_1 - F_0 - F_2}$

$$\text{Mode} = 30 + \frac{(40-30)(31-26)}{2\times 31 - 26 - 16}$$

$$\text{Mode} = 30 + \frac{50}{20}$$

Mode = 30 + 2.5
Mode = 32.5

60. 20 homes from a local area by a group of students household family members in a survey conducted on of the following frequency table related to the number of family numbers.

Size of family	Family numbers
1–3	7
3–5	9
5–7	2
7–9	1
9–11	1

Find the mode of the given data.

(a) 3.571 (b) 3.444
(c) 3.628 (d) 3.286

RRB Group-D – 20/09/2018 (Shift-III)

Ans : (b)

Size of family	Family numbers
1–3	$7 = f_0$
3–5	$9 = f_1$
5–7	$2 = f_2$
7–9	1
9–11	1

The frequency of category 3-5 is the highest so polymer category is 3-5
$L = 3, f_0 = 7, f_1 = 9, f_2 = 2$ and $h = 2$

$$\text{mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right)\times h$$

$$= 3 + \frac{9-7}{18-7-2}\times 2$$

$$= 3 + \frac{2}{9}\times 2 = \frac{31}{9} = 3.444$$

61. Wickets taken by a bowler in 12 cricket matches are as follows:
2, 6, 4, 3, 5, 0, 3, 2, 1, 3, 2, 3
What will be the mode of the given data?

(a) 4 (b) 2
(c) 3 (d) 1

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (c) : Polymer means that the number has come more often or has a higher frequency.
Therefore, the given polymer has a high frequency of 3
$\therefore$ Polymer of figures = 3

62. What will be the mode of the given following data?
25, 45, 58, 87, 45, 54, 65, 12, 25, 59, 42, 60

(a) 25 (b) 45
(c) 45, 54 (d) 45, 25

RRB Group-D – 09/10/2018 (Shift-I)

Ans. (d) : given figures– 25, 45, 58, 87, 45, 54, 65, 12, 25, 59, 42, 60

It has a maximum frequency of 25 (twice) and 45 (twice) also.

So Required mode is 25, 45

63. Details of number of people taking loans from bank based on their age group interval given below Find the mode?

age group	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
No. of people	37	38	70	42	13

Find the mode.

(a) 45.33 (b) 44.89

(c) 45.67 (d) 45.12

RRB Group-D – 15/11/2018 (Shift-III)

Ans : (a) :

Age category	Age category persons number
20 - 30	37
30 - 40	38
40 - 50	70 mode category
50 - 60	42
60 - 70	13

here - L = minimum limit of mode category = 40

f_1 = number of mode category = 70

f_0 = Number of the persons belongs to upper cless of polymer class = 38

f_2 = Number of persons belongs to the class below the polymer class = 42

i = higher limit – lower limit (quadratic) = 10

$$\text{mode (z)} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

$$= 40 + \frac{70-38}{70\times 2 - 38 - 42} \times 10$$

$$= 40 + \frac{32}{140-80} \times 10$$

$$= 40 + \frac{320}{60} = 40 + 5.33 = 45.33$$

64. Below, details of 100 students present in the class is given on the basis of ther presence (day).

The number of days of presence	6-10	10-14	14-18	18-22	22-26
Number of students	9	28	34	18	11

What will be the mode of the data?

(a) 15.09 (b) 15.01

(c) 14.71 (d) 15.04

RRB Group-D – 12/11/2018 (Shift-I)

Ans. (a) :

Number	**frequncy**
6-10	9
10-14	$28 = f_0$
14-28	$34-f_1$
18-22	$18 = f_2$
22-26	11

$$z\,(\text{Mode}) = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

L = 14 $f_1 = 34$ $f_0 = 28$

$f_2 = 18$ i = 4

$$z = 14 + \frac{34-28}{68-28-18} \times 4 = 14 + \frac{24}{22}$$

z = 15.09

65. What will be the mode of the given data?

12, 1, 10, 1, 9, 3, 4, 9, 7, 9

(a) 9 (b) 12

(c) 1 (d) 7

RRB NTPC 17.01.2017 Shift-1

Ans : (a) In the given data the mode = 9 (3 times)

∵ mode is the highest frequency number.

66. What will be the mode of the given data as follows– 32, 34, 35, 36, 35, 34, 33, 35, 33, 31 and 37

(a) 33 (b) 34

(c) 35 (d) 32

RRB NTPC 04.04.2016 Shift : 3

Ans :(c) 32, 34, 35, 36, 35, 34, 33, 35, 33, 31, 37

∵ highest frequence is 3 of the number 35.

So mode = 35

67. What will be the mode of the given data given as – 12, 14, 15, 16, 15, 14, 13, 15, 13, 11 and 17

(a) 13 (b) 14

(c) 15 (d) 12

RRB NTPC 03.04.2016 Shift : 3

Ans :(c) If the frequency of a number is the most frequent in the data, then that number is the mode of the given data.

So mode of data = 15

68. What will be the mode of the given data given as– 2, 4, 5, 6, 5, 4, 3, 5, 3, 1 and 7

(a) 3 (b) 4

(c) 5 (d) 2

RRB NTPC 02.04.2016 Shift : 3

Ans : (c) 2, 4, 5, 6, 5, 4, 3, 5, 3, 1, 7

Frecquency of 5 is highest in the given data.

∴ mode = 5

69. If the mode of the given data is 52, then find the value of x?

52, 45, 49, 54, 56, x-3, 56

(a) 52 (b) 55

(c) 54 (d) 56

RRB NTPC 29.03.2016 Shift : 2

Ans : (b) $\because$ Mode of figures = 52

$\therefore x - 3 = 52 \Rightarrow \boxed{x = 55}$

70. What will be the mode of the given data–

$\mathbf{1, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{1}{4}, 2, \frac{1}{2}, \frac{1}{4}, \frac{2}{4}}$

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$

(c) $\frac{3}{4}$ (d) 1

RRB NTPC 07.04.2016 Shift : 3

Ans :(b)

$\because$ frequency of $\frac{1}{2}$ is maximum (3) in data.

$\therefore$ mode $= \frac{1}{2}$

Type - 4

71. Find the standard deviation of {11, 7, 10, 13, 9}.

(a) 1 (b) 2

(c) 4 (d) 5

RRB RPF SI – 10/01/2019 (Shift-II)

Ans : (b) Standard deviation $= \sqrt{\frac{\sum(x-\bar{x})^2}{n}}$

Where

$\bar{x} \rightarrow$ mean

$n \rightarrow$ The number of terms

$\bar{x} = \frac{11+7+10+13+9}{5} = \frac{50}{5} = 10$

$\sum (x-\bar{x})^2 = (11-10)^2 + (7-10)^2 + (10-10)^2 + (13-10)^2 + (9-10)^2 = 1 + 9 + 0 + 9 + 1 = 20$

Standard deviation $= \sqrt{\frac{20}{5}} = 2$

72. The value of the 5 variance is 16. If each value becomes double then find the new value of standard deviation?

(a) 16 (b) 4

(c) 10 (d) 8

RRB JE - 28/05/2019 (Shift-I)

Ans : (d) We know that-

Standard deviation $= \sqrt{\text{Variance}}$

Standard deviation $= 4$

Note : if the figures are added subtracted, multiplied or divided by a given number, then the standard deviation is the same process.

Hence the new standard deviation $= 4 \times 2 = 8$

73. The variance of the 5 values is 36. If each value becomes double, then what will be the standard deviation?

(a) 12 (b) 6

(c) 18 (d) 10

RRB JE - 30/05/2019 (Shift-III)

Ans : (a) Variance $= \sigma^2$

Standard deviation $= \sqrt{\sigma^2} = \sqrt{36}$

Standard deviation $= \sigma = 6$

New standard deviation $= \lambda\sigma$

(Where λ = n times each value)

$= 2 \times 6$

$= 12$

74. What will be the standard deviation of the given following data–

6, 12, 9, 7, 8, 4, 3, 12, 15, 4

(a) 3.80 (b) 2

(c) 3.48 (d) 4

RRB JE - 26/06/2019 (Shift-I)

Ans. (a)

x	6	12	9	7	8	4	3	12	15	4
$d = (x-\bar{x})$	–2	+4	1	–1	0	–4	–5	+4	+7	–4
d^2	4	16	1	1	0	16	25	16	49	16

$\Sigma d^2 = 144$

Mean $(\bar{x}) = \frac{6+12+9+7+8+4+3+12+15+4}{10} = 8$

Standard deviation $= \sqrt{\frac{\Sigma d^2}{n}} = \sqrt{\frac{144}{10}} = \sqrt{14.4} = 3.8$

75. What will be the standard deviation of the first n natural number?

(a) $\sqrt{\frac{n^2-1}{12}}$ (b) $\frac{n(2n+1)}{3}$

(c) $\sqrt{\frac{n^2+1}{6}}$ (d) $\frac{n(n+1)}{12}$

RRB JE - 02/06/2019 (Shift-II)

Ans. (a) σ (S.D) = $\sqrt{\frac{(\Sigma x_i^2)}{n} - \left(\frac{\Sigma x_i}{n}\right)^2}$

First 'n' natural number

$\Sigma x_i = 1 + 2 + 3 + ----- + n = \frac{n(n+1)}{2}$

$(\Sigma x_i)^2 = (1^2 + 2^2 + 3^2 + ----- + n^2) = \frac{n(n+1)(2n+1)}{6}$

$\sigma = \sqrt{\frac{n(n+1)(2n+1)}{6 \times n} - \left(\frac{n(n+1)}{2n}\right)^2}$

$\sigma = \sqrt{\frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}}$

$\sigma = \sqrt{(n+1)\frac{(4n+2-3n-3)}{12}}$

$\sigma = \sqrt{\frac{(n+1)(n-1)}{12}} = \sqrt{\frac{n^2-1}{12}}$

76. Sachin Tendulkar scored run in ten innings agains Australia 38, 70, 48, 34, 42, 55, 63, 46, 54, and 44. Find the average deviation with respect to mean.

(a) $\frac{44}{5}$ (b) $\frac{43}{5}$

(c) $\frac{41}{5}$ (d) $\frac{42}{5}$

RRB Group-D – 05/12/2018 (Shift-III)

Ans : (b)

average = $\frac{34+38+42+44+46+48+54+55+63+70}{10}$

$= \frac{494}{10} = \frac{247}{5}$

mean= $\underbrace{34+38+42+44+46}_{\frac{204}{5}}$ $\underbrace{48+54+55+63+70}_{\frac{290}{5}}$

Standard deviation $= \frac{247}{5} - \frac{204}{5} = \frac{43}{5}$

77. If the standard deviation of the given numbers is 255. Then find the value of 'd'

1, 1 + d, 1 + 2d.....1+ 100d

(a) 20.2 (b) 10.1

(c) 20.0 (d) 10.5

RRB Group-D – 16/11/2018 (Shift-II)

Ans : (b) 1,1 + d, 1 + 2d.....1+ 100 d is in arithmetic progression.

$\because$ Number of total terms = 101

Arithmatic mean $(\bar{x}) = \frac{1+(1+d)+.....(1+100d)}{101}$

$= \frac{101(1+50d)}{101}$

$\bar{x} = 1+50d$

mean deviation $= \frac{1}{101}\sum_{i=0}^{100}|x_i - \bar{x}|$

$= \frac{1}{101}(|-50d|+|-49d|+....|-d|+0+|d|+|2d|+.....+|50d|)$

$255 = \frac{2d}{101}\frac{(50\times51)}{2}$

$255 = \frac{d}{101}\times50\times51$

$d = \frac{255\times101}{50\times51} = \frac{25755}{2550} = 10.1$

78. What will be the standard deviation of the gtiven data–3, 10, 10, 4, 7, 10, 5

(a) $\frac{49}{7}$ (b) $\frac{19}{7}$

(c) $\frac{50}{7}$ (d) $\frac{18}{7}$

RRB NTPC 17.01.2017 Shift-1

Ans : (d) Data 3, 10, 10, 4, 7, 10, 5

Mean $= \frac{3+10+10+4+7+10+5}{7} = \frac{49}{7}$

Mean = 7

Mean deviation $= \sum_{i=1}^{N}\frac{|M-Xi|}{N}$

$= \frac{|7-3|+|7-10|+|7-10|+|7-4|+|7-7|+|7-10|+|7-5|}{7}$

$= \frac{4+3+3+3+0+3+2}{7}$

$\boxed{\text{mean deviation} = \frac{18}{7}}$

79. What will be the standard deviation of the given set {10, 11, 12, 9, 8}

(a) 1 (b) $\sqrt{2}$

(c) 2 (d) $2\sqrt{2}$

RRB NTPC 31.03.2016 Shift : 2

Ans : (b) $\bar{x} = \frac{10+11+12+9+8}{5} = \frac{50}{5} = 10$

$\sum_{i=1}^{5}(x_i - \bar{x})^2 = (10-10)^2 + (11-10)^2 + (12-10)^2 + (9-10)^2 + (8-10)^2$

$= 0 + 1 + 4 + 1 + 4 = 10$

$\therefore$ Standard deviation $= \sqrt{\frac{\sum_{i=1}^{5}(x_i - \bar{x})^2}{N}} = \sqrt{\frac{10}{5}} = \sqrt{2}$

80. If the standard deviation of a distribution is 6, then what is the value of variance?

(a) 8 (b) 24
(c) 36 (d) 12

RRB NTPC 18.01.2017 Shift : 3

Ans : (c) Variance = (Standard deviation)^2
$= (6)^2 = 36$

81. If the variance of a data set is 196, the standard deviation will be?

(a) ±14 (b) 14
(c) 96 (d) 98

RRB NTPC 11.04.2016 Shift : 1

Ans : (b) Standard deviation
$= \sqrt{\text{variance}} = \sqrt{196} = 14$

82. If the variance of data set is 121, then find the standard deviation of the data.

(a) ±11 (b) 11
(c) 21 (d) 60.5

RRB NTPC 11.04.2016 Shift : 2

Ans : (b) Standard deviation of data $= \sqrt{\text{variance}}$
$= \sqrt{121} = 11$

83. If the variance of data set is 81, then find the standard deviation of the data?

(a) ±9 (b) 9
(c) 81 (d) 40.5

RRB NTPC 11.04.2016 Shift : 3

Ans : (b) Standard deviation of data $= \sqrt{81}$
$= \pm 9 = 9$

84. The coefficiant of a set of data is 64, then find out the standard deviation?

(a) ±8 (b) 8
(c) 14 (d) 64

RRB NTPC 07.04.2016 Shift : 1

Ans : (b)
Standard deviation $= \sqrt{\text{coefficient}} = \sqrt{64} = \pm 8 = 8$

85. The variance of a data set is 361, then find out the value of standard deviation?

(a) ±19 (b) 19
(c) 361 (d) 180.5

RRB NTPC 07.04.2016 Shift : 3

Ans :(b) Standard deviation $= \sqrt{\text{variance}}$
$= \sqrt{361}$
$= \pm 19$
Since the standard deviation is not negative
So standard deviation = 19

86. The variance of a data set is 169, then what will be the standard deviation?

(a) ±13 (b) 13
(c) 69 (d) 845

RRB NTPC 26.04.2016 Shift : 2

Ans : (b) Standard deviation $= \sqrt{\text{variance}}$
$= \sqrt{169}$
$= \pm 13$
$= 13$

87. In n observations $x_1, x_2, x_3, ...x_n$, frequency will be given as $f_1, f_2, f_3 .....f_x$. What will be the standard deviation $\bar{x}$

(a) $\sqrt{\frac{\sum_{i=1}^{n} f_i(x_i - \bar{x})}{\sum_{i=1}^{n} f_i}}$ (b) $\sqrt{\frac{\sum_{i=1}^{n} f_i(x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i}}$

(c) $\sqrt{\frac{\sum_{i=1}^{n} f_i(x_i^2 - \bar{x})}{\sum_{i=1}^{n} f_i}}$ (d) $\sqrt{\sum_{i=1}^{n} f_i(x_i - \bar{x})}$

RRB NTPC 26.04.2016 Shift : 3

Ans : (b) Standard deviation $= \sqrt{\frac{\sum_{i=1}^{n} f_i(x_i - \bar{x})^2}{\sum_{i=1}^{n} f_i}}$

where $\bar{x}$ = mean

88. Mean of given observation $x_1, x_2, x_3, ...x_n$ is $\bar{x}$. What will be the standard deviation of the n observation?

(a) $\sqrt{\frac{\sum_{1}^{n}(x_i - \bar{x})}{n}}$ (b) $\sqrt{\frac{\sum_{1}^{n}(x_i - \bar{x})^2}{n}}$

(c) $\sqrt{\frac{\sum_{1}^{n}(x_i^2-\bar{x})}{n}}$ (d) $\sqrt{\frac{\sum_{1}^{n}(x_i-\bar{x})^2}{n}}$

RRB NTPC 30.04.2016 Shift : 3

Ans : (b) Standard deviation $(S.D.)=\sqrt{\frac{\sum_{1}^{n}(x_i-\bar{x})^2}{n}}$

where $\sum_{1}^{n}(x_1-\bar{x})^2$ = variables

Type - 5

89. When a number is chosen at random, what will be the probabily of two digit numbers, but number not a co-prime?

(a) 7/30 (b) 23/30

(c) 21/90 (d) 67/90

RRB JE - 29/05/2019 (Shift-II)

Ans : (b) Total number of two digits = 90

Total prime numbers of two digits = 21

Total composite numbers of two digits = 69

$\therefore$

Probability $=\frac{\text{number of favorable results}}{\text{number of total results}}=\frac{69}{90}=\frac{23}{30}$

90. When a pair of dice is thrown, what is the probability of the sum of numbers being odd?

(a) 1 (b) 0.25

(c) 0.4 (d) 0.5

RRB JE - 29/05/2019 (Shift-II)

Ans : (d) When two dice are thrown,

The probability of occurrence of some event is n (s) = 36

The probability of odd sum of numbers = n (E) = 18

$\therefore$ probability of adding odd = $\frac{n(E)}{n(S)}=\frac{18}{36}=\frac{1}{2}=0.5$

91. Satish puts 5 yellows and 3 blue balls in a closed box. His brother Manish picks two balls at random. Calculate the probability that balls picked are of the same colour.

(a) $\frac{15}{28}$ (b) $\frac{15}{23}$

(c) $\frac{13}{28}$ (d) $\frac{11}{23}$

RRB RPF Constable – 17/01/2019 (Shift-III)

Ans : (c) Number of total balls = 8

Probability of picking balls of the same colour

$$=\frac{{}^5C_2+{}^3C_2}{{}^8C_2} \qquad \left[{}^nc_r=\frac{n!}{r!(n-r)!}\right]$$

$$=\frac{\frac{5!}{3!\times 2!}+\frac{3!}{1!\times 2!}}{\frac{8!}{2!\times 6!}}$$

$$\frac{\frac{5\times 4\times 3!}{2\times 1\times 3!}+\frac{3\times 2!}{2!\times 1}}{\frac{8\times 7\times 6!}{2\times 1\times 6!}}$$

$$=\frac{\frac{5\times 4}{2\times 1}+3}{\frac{8\times 7}{2\times 1}}$$

$$=\frac{\frac{5\times 4}{2}+3}{8\times 7/2}=\frac{\frac{26}{2}}{\frac{56}{2}}=\frac{13}{28}$$

92. In a shooting competition, to hit a target, probability 1/2 for A, 2/3 for B and 3/4 for C. If they hit the target together, then what will be the probability of penetration for each of them?

(a) 1/6 (b) 3/8

(c) 2/3 (d) 1/4

RRB JE - 28/05/2019 (Shift-I)

Ans : (d) Probability of A goal (A) = $\frac{1}{2}$

And the probability of not penetrating the goals of A

(A') = $1-\frac{1}{2}=\frac{1}{2}$

Probability of B goal (B) = $\frac{2}{3}$

And probability of not hitting B goal (B') = $1-\frac{2}{3}=\frac{1}{3}$

Probability of C's target (C) = $\frac{3}{4}$

and probability of not hitting the goals of

(C') = $1-\frac{3}{4}=\frac{1}{4}$

Probability of hitting the target of one when B and C are Hitting simultaneously

= (A×B'×C')+(A'×B×C') + (A'×B'×C)

$$=\frac{1}{2}\times\frac{1}{3}\times\frac{1}{4}+\frac{1}{2}\times\frac{2}{3}\times\frac{1}{4}+\frac{1}{2}\times\frac{1}{3}\times\frac{3}{4}$$

$$=\frac{1}{24}+\frac{2}{24}+\frac{3}{24}=\frac{6}{24}=\frac{1}{4}$$

93. One black, one red and one green dice are thrown together, what is the probability of sum of three numbers is ≥17

(a) 7/216 (b) 5/216
(c) 1/54 (d) 1/36

RRB JE - 30/05/2019 (Shift-III)

Ans : (c) The probability of the sum of the numbers exceeding from 17 or 17 when throwing all three passes together—

(5, 6, 6), (6, 5, 6), (6, 6, 5), (6,6,6)

Total probability $= 6 \times 6 \times 6 = 216$

$\therefore$ Intended probability $= \frac{4}{216} = \frac{1}{54}$

94. Find the probability of the sum being 10, when a dice are thrown two times randemely.

(a) 3 (b) 1/36
(c) 1/12 (d) 5/36

RRB RPF SI – 05/01/2019 (Shift-II)

Ans : (c) 2 times a dice is thrown—

Total probability $N(E) = 6^2 = 36$

The probability of the sum of digits being 10 is N(S) = (4, 6) (6, 4) (5, 5) = 3

So intended probability $P(E) = \frac{N(S)}{N(E)}$

$= \frac{3}{36} = \frac{1}{12}$

95. What will be the probability to remove face card from card dick?

(a) $\frac{6}{13}$ (b) $\frac{12}{13}$
(c) $\frac{3}{13}$ (d) $\frac{3}{26}$

RRB Group-D – 12/10/2018 (Shift-I)

Ans. (c) : Total number of cards = 52

Number of face card = 12

so probability of face card $= \frac{12}{52} = \frac{3}{13}$

96. To ace out the deck of cards probability can be

(a) $\frac{12}{13}$ (b) $\frac{15}{26}$
(c) $\frac{9}{13}$ (d) $\frac{1}{13}$

RRB Group-D – 16/10/2018 (Shift-I)

Ans. (d) : The number of card is 52.

The number of Ace in deck of card is 4.

Probability of getting ace out of the deck of cards

$= \frac{4}{52} = \frac{1}{13}$

97. A box contains 100 pens, out of which eight are defective. One pen is out from the box. Find the probability that the pen is not failse.

(a) 23/25 (b) 8/100
(c) 100/8 (d) 25/23

RRB Group-D – 29/10/2018 (Shift-III)

Ans : (a)

Number of total pens = 100

Number of waste pens = 8

Probability of taking waste pens $= \frac{8}{100}$

Probability of not having waste pen

$= 1 - \frac{8}{100} = \frac{92}{100} = \frac{23}{25}$

98. If a box contians 3 white cushions, 4 red cushions and 5 blue cushions. What is the probability of selection of one white or blue cushion?

(a) 2/3 (b) 3/4
(c) 1/4 (d) 1/9

RRB NTPC 28.04.2016 Shift : 2

Ans : (a) White cushion = 3
red cushion = 4
blue cushion = 5

$$\boxed{\text{Probability} = \frac{\text{favorable events}}{\text{total events}}}$$

Total events $= {}^{12}C_1$

favorable probabilities to choose 1 white or 1 blue custion $= {}^3C_1 + {}^5C_1$

So probability $= \frac{{}^3C_1 + {}^5C_1}{{}^{12}C_1}$ $\left[{}^nC_r = \frac{\lfloor n}{\lfloor r \lfloor n-r}\right]$

$$= \frac{\frac{\lfloor 3}{\lfloor 1 \lfloor 3-1} + \frac{\lfloor 5}{\lfloor 1 \lfloor 5-1}}{\frac{\lfloor 12}{\lfloor 1 \lfloor 12-1}}$$

$$= \frac{\frac{\lfloor 3}{\lfloor 2} + \frac{\lfloor 5}{\lfloor 4}}{\frac{\lfloor 12}{\lfloor 11}} = \frac{\frac{3\times\lfloor 2}{\lfloor 2} + \frac{5\times\lfloor 4}{\lfloor 4}}{\frac{12\times\lfloor 11}{\lfloor 11}} = \frac{3+5}{12} = \frac{8}{12} = \frac{2}{3}$$

Note :- use $\boxed{+}$ for 'or' and $\boxed{\times}$ for 'and'.

99. When a coin is tossed once, what are the prabability of coming Head?

(a) 1 (b) 1/2
(c) 2 (d) Zero

RRB NTPC 29.04.2016 Shift : 1

Ans : (b) Head favorable probability when a coin is tossed

Required probability $= \frac{\text{Favourable probability}}{\text{total probability}} = \frac{1}{2}$

So probability to come Head $= \frac{1}{2}$

Type - 5

100. Find the range of data - 9, 5, 9, 3, 4, 7, 8, 4, 8, 9, 5, 9

(a) 3 (b) 5
(c) 6 (d) 4

RRB RPF Constable – 24/01/2019 (Shift-III)

Ans : (c) Data range = highest value − lowest value
$= 9 - 3 = 6$

101. Find the range of data-11, 13, 9, 17, 13, 19, 10, 11

(a) 6 (b) 10
(c) 11 (d) 13

RRB RPF SI – 12/01/2019 (Shift-II)

Ans : (b) Data range = highest value – lowest value
$= 19 - 9 = 10$

102. The standard diviation of a group of values is 4.5. If each value increases by K. then find the variance of group of new values?

(a) 10.5 (b) 20.25
(c) 100.25 (d) 4.5

RRB JE - 26/06/2019 (Shift-I)

Ans. (b) If in each value is increased by K, the standard deviation will have no effect.
Variance = (Standard deviation)^2
$= (4.5)^2 = 20.25$

103. The variance of a set of values X_1, X_2, ... X_n by which of the following formula has given?

(a) $\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2$

(b) $\left(\frac{\Sigma x}{n}\right)^2 - \frac{\Sigma x^2}{n}$

(c) $\left(\frac{\Sigma x}{n}\right)^2$

(d) $\frac{\Sigma x^2}{n} - \frac{\Sigma x}{n}$

RRB JE - 01/06/2019 (Shift-II)

Ans : (a) Variance (σ^2) = $\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2$

104. What will be the range of first 7 prime number?

(a) 15 (b) 8.3
(c) 9 (d) 17

RRB JE - 27/06/2019 (Shift-III)

Ans : (a) First 7 prime number= 2, 3, 5, 7, 11, 13, 17
Range = maximum number – minimum number
Range = 17 – 2 = 15

105. If the upper quartile represents 75% of the data and if the lower quartile represents 25% of the data, then which will show the median?

(a) 20% (b) 100%
(c) 50% (d) 80%

RRB Group-D – 05/12/2018 (Shift-III)

Ans : (c) Median = $\frac{75\% + 25\%}{2} = \frac{100\%}{2} = 50\%$

106. A set of four numbers. The mean of these three smallest numbers is 19 and the mean of three larger numbers is 23. What will be the range of this set?

(a) 18 (b) 12
(c) 14 (d) 15

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (b) Sum of the three smallest number in four numbers = 19 × 3 = 57
sum of three smallest numbers = 23 × 3 = 69
Range of sets = 69 – 57 = 12

107. The mean of three numbers is 20. The range of this data set is 12, while the difference between two smallest numbers is 3. Find the larger number.

(a) 28 (b) 25
(c) 27 (d) 24

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (c) : Suppose numbers = x, y, z where x < y < z
As per the question-
$x + y + z = 60 - (i)$
$z - x = 12 \Rightarrow z = 12 + x$ .....(ii)
$y - x = 3 \Rightarrow y = 3 + x$ .....(iii)
to solving of equation (i),(ii) and (iii)
x + y + z = 60
x + 3 + x + 12 + x = 60
3x + 15 = 60
3x = 45
x = 15
y = 18
z = 12 + x
z = 12 + 15
z = 27
$\boxed{z = 27}$ highest number (z) = 27

108. The mean of three numbers is 33 and the range of data is 29. The middle number is 27 more than the sum of the other two numbers. What is the largest number among these three numbers?

(a) 46 (b) 45
(c) 48 (d) 47

RRB Group-D – 03/12/2018 (Shift-II)

Ans : (a) Suppose three number is x > y > z
as per question—

$$\frac{x+y+z}{3} = 33$$

x + y + z = 99 ----------(I)
x – z = 29 ----------(II)
and y = x + z – 27
y + 27 = x + z ----------(III)
from equation (I) and (III) -
y + y + 27 = 99
2y = 72,
y = 36

Putting the value of y in equation (III)
y + 27 = x + z
36 + 27 = x + z
x + z = 63 ----------(IV)
equation (II) + (IV)
2x = 92,
x = 46
'Putting the value of x in equation (IV)-
z = 63 – 46
z = 17
So number x, y and z is 46, 36, 17
So largest numbers = 46

109. The mean of three numbers 35 and the range of data was 24. Difference between largest and middle number is equal to three times of the difference between the smallest and the middle number. What are the largest of these three numbers?

(a) 51 (b) 50
(c) 49 (d) 52

RRB Group-D – 23/10/2018 (Shift-III)

Ans : (c) Suppose numbers is as a, b, c where a> b> c

$$\text{Mean} = \frac{a+b+c}{3} = 35$$

a+b+c = 105 .....(1)
range a-c = 24 .....(2)
and a-b = 3(b-c) .....(3)
where a-b = 3b -3c
a+3c = 4b
a + c = 4b-2c .....(4)
Putting equation (4) to the value of a + c in equation (1)
b+ 4b-2c=105
5b-2c= 105.....(5)
Putting equation (2) to a = 24 + c in equation (1)
24+c + b + c =105
b+2c = 81 .....(6)
Solving equation (5) and (6)
b = 31 c= 25
from equation (1)
a+31+25= 105
a + 56 = 105
$\boxed{a = 49}$
Hence, the largest number = a = 49

110. What will be the range of the following data?
6, 7, 8, 9, 5, 6, 7, 4, 8, 9, 5, 9

(a) 2 (b) 3
(c) 4 (d) 5

RRB NTPC 31.03.2016 Shift : 1

Ans : (d) lowest limit = 4
highest limit = 9
∴ highest limit – lowest limit = 9 – 4 = 5

111. What will be the range of the following data?
2, 1, 2, 3, 4, 5, 7, 3, 5, 2, 4

(a) 5 (b) 4
(c) 3 (d) 6

RRB NTPC 31.03.2016 Shift : 2

Ans : (d) lowest limit = 1
highest limit = 7
Range = highest limit – lowest limit
∴ Range = 7 – 1 = 6

112. What will be the range of the following data?
12, 11, 18, 28, 19, 13, 19, 18

(a) 11 (b) 17
(c) 18 (d) 19

RRB NTPC 30.03.2016 Shift : 2

Ans : (b) The difference between the highest and lowest values of the given data is called of the range.
Range = 28–11 = 17

113. If the standard deviation of a population is 9.5, then what will be its variance?

(a) 19 (b) 90.25
(c) 81.25 (d) 93.25

RRB NTPC 19.04.2016 Shift : 1

Ans : (b) Variance = $(\text{Standard deviation})^2$
$= (9.5)^2$
= 90.25

114. If the standand deviation of a population is 4.5, then what will be its variance?

(a) 20.25 (b) 20
(c) 9 (d) 18

RRB NTPC 16.04.2016 Shift : 1

Ans : (a) Variance = $(\text{Standard deviation})^2$
$= (4.5)^2 = 20.25$

115. If the variance of data 2, 4, 5, 6, 8, 18 is 23.33 then what will be the variance of the given following data 4, 8, 10, 12, 16, 36

(a) 11.66 (b) 46.66
(c) 93.3333 (d) 483

RRB NTPC 11.04.2016 Shift : 1

Ans : (b) ∴ Data numbers have doubled, so their variance will also be doubled.
∴ Variance = 2 × 23.33
= 46.66

116. What will be the range of the following data? 10, 21, 5, 1, 3, 17, 19, 2

(a) 19 (b) 10
(c) 20 (d) 17

RRB NTPC 09.04.2016 Shift : 3

Ans : (c) Range of numbers = high limit – low limit
= 21 – 1 = 20

117. The following information is $60\sum x^2 = 18000$, $\sum x = 960$ variance is related to the size of a sample.

(a) 55 (b) 44
(c) 22 (d) 16

RRB NTPC 07.04.2016 Shift : 3

Ans :(b) $60\sum x^2 = 18000$

$\sum x^2 = \frac{18000}{60} \Rightarrow \boxed{\sum x^2 = 300}$

$\because \quad \sum x = 960$

average of 60 terms $= \frac{960}{60} = 16$

$\sum x^2 = (16)^2 \Rightarrow \boxed{\sum x^2 = 256}$

∴ Variance = 300 – 256 = 44

118. If the standard deviation of the population is 11, then what will be the variance of population?

(a) 44 (b) 121
(c) 22 (d) 33

RRB NTPC 22.04.2016 Shift : 1

Ans : (b) Standard deviation of population = 11
Population variance (standard deviation)^2=$(11)^2$ = 121

119. What will be the range of the following data? 3, 1, 4, 6, 5, 7, 3, 8, 1, 4

(a) 3 (b) 8
(c) 7 (d) 6

RRB NTPC 26.04.2016 Shift : 1

Ans : (c) writing in given increasing
1, 1, 3, 3, 4, 4, 5, 6, 7, 8
Data rang = large number – small number
= 8 – 1 = 7

120. If the standard deviation of a distribution is 4 then what is the value of variance?

(a) 8 (b) 9
(c) 16 (d) 12

RRB NTPC 28.04.2016 Shift : 3

Ans : (c) Standard deviation = 4
∴ Variance = (Standard deviation)^2 = $(4)^2$ = 16

Type - 7

121. The mean of a distribution is 24 and the standard deviation is 6. Then what is the value of variance cofficient.

(a) 50% (b) 25%
(c) 100% (d) 75%

RRB RPF SI – 11/01/2019 (Shift-I)

Ans : (b) Variance coefficient $= \frac{\text{standard deviation}}{\text{mean}}$

$= \frac{6}{24} \times 100 = 25\%$

122. If the standard deviation of the population is 10, then what will be the variance coefficient?

(a) 100 (b) 30
(c) 5 (d) 20

RRB RPF Constable – 25/01/2019 (Shift-I)

Ans : (a)
Standrad deviation of population= 10
So variance coefficient = (standard deviation)^2
= $(10)^2$ = 100

123. The mean of the distribution is 14 and the standard deviation is 5. Find the variance coefficient.

(a) 60.4% (b) 27.9%
(c) 35.7% (d) 48.3%

RRB JE - 01/06/2019 (Shift-I)

Ans : (c) Variance coefficient $= \frac{\text{standard deviation}}{\text{mean}}$

$= \frac{5}{14} \times 100$

$= \frac{500}{14} = 35.7\%$

124. The ratio of number of blue and red balls in a bag is constant. When the number of red balls are 68 the number of blue balls was 36. Number of blue balls being 63, what is the number of red balls in the bag?

(a) 119 (b) 98
(c) 110 (d) 102

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (a) blue : red = 36 : 68

= 9: 17

When blue balls number is 63 then suppose red balls numbers is n then-

$$\frac{63}{n} = \frac{9}{17}$$

$9n = 63 \times 17$

$$n = \frac{63 \times 17}{9}$$

$n = 7 \times 17$

$n = 119$

So red balls number = 119

125. In the given following data find out the LCM of mode, median and mean?

7, 2, 10, 4, 3, 12, 8, 4, 6, 4.

(a) 30 (b) 20

(c) 12 (d) 60

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (d) 7, 2, 10, 4, 3, 12, 8, 4, 6, 4

Writing in ascending order- 2, 3, 4, 4, 4, 6, 7, 8, 10, 12

⇒ Mode = 4 (the highest frequency)

$$\text{Median} = \frac{1}{2}\left[\frac{n^{th}}{2} + \left(\frac{n}{2}+1\right)^{th}\right]$$

$$\text{Median} = \frac{1}{2}\left(5^{th} + 6^{th}\right)\text{term} \begin{cases}\text{where n = terms number} \\ n = 10\end{cases}$$

$$= \frac{1}{2}(4+6) = 5 = \frac{1}{2} \times 10 = 5$$

$$\text{Mean} = \frac{\text{total sums}}{\text{total number}} = \frac{2+3+4+4+4+6+7+8+10+12}{10}$$

$$\text{mean} = \frac{60}{10} = 6$$

LCM of 4, 5, 6 = 2 × 2 × 3 × 5 = 60

126. For the following data what is the (mode×median+mean)

9, 1, 11, 3, 2, 12, 8, 4, 6, 4.

(a) 34 (b) 31

(c) 26 (d) 29

RRB Group-D – 30/10/2018 (Shift-I)

Ans : (c) Mode = 4(maximum)

writing in ascending order-

1, 2, 3, 4, 4, 6, 8, 9, 11, 12

$$\text{mean} = \frac{1+2+3+4+4+6+8+9+11+12}{10}$$

$$= \frac{60}{10} = 6$$

Total number(n) = 10 (even)

$$\text{median} = \frac{\frac{n}{2}\text{terms value} + \left(\frac{n}{2}+1\right)\text{terms value}}{2}$$

$$= \frac{5^{th}\text{ terms value} + 6^{th}\text{ terms value}}{2}$$

$$= \frac{4+6}{2} = \frac{10}{2} = 5$$

= mode × median + mean

= (4 × 5) + 6 = 26

127. If the mean and median of the numbers 3, 4, a, b, 10 are 6 and 5 respectively where a>b, What will be the value of a and b.

(a) 7,6 (b) 5,8

(c) 6,7 (d) 8,5

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (b) : mean of 3,4,a,b, 10 is 6 and median 5 a < b

$$\text{Mean} = \frac{\text{sum of total number}}{\text{total number}}$$

$$6 = \frac{3+4+a+b+10}{5}$$

30 = 17 + a + b

a + b = 13

median is 5–

median when terms number are odd– $\left(\frac{n+1}{2}\right)$

terms number(n) = 5

$$\text{median} = \left(\frac{5+1}{2}\right)^{th}\text{ term} = 3^{rd}\text{ term}$$

5 = a

a + b = 13

5 + b = 13

b = 8

a = 5, b = 8

128. The mean of the three numbers was 15 and the range of figure was 9. Difference between larges number and middle number is equals to two times of the difference between middle number and smallest number what will be the largest number in the given three numbers?

(a) 20 (b) 21

(c) 19 (d) 22

RRB Group-D – 22/10/2018 (Shift-II)

Ans : (a) Suppose three numbers is a, b and c

where a < b < c

$$\frac{a+b+c}{3} = 15$$

$$\Rightarrow a+b+c = 45 - - - - - (I)$$

then $c - a = 9$ --------(II)

and $c - b = 2(b - a)$

$\Rightarrow c - b = 2b - 2a$

$\Rightarrow 3b = 2a + c \Rightarrow b = \frac{2a+c}{3}$ ----(III)

Putting 'b' value in equation (I) from equation (III)—

$$a + \left(\frac{2a+c}{3}\right) + c = 45$$

$$\Rightarrow 5a + 4c = 135 \text{------(IV)}$$

from equation (IV) + equation (II) × 5—

$9c = 180 \Rightarrow c = 20$

from equation (II)

$20 - a = 9 \Rightarrow a = 11$

from equation (III) - $b = \frac{2\times 11 + 20}{3} = 14$

So largest number = 20

129. The four integers a, b, c and d arranged in ascending order, the range of the set is 20. The difference between c and a is equal to the difference between d and b. The integral mean of the numbers is 25. In the following what will be the value of a?

(a) 13 (b) 15

(c) 14 (d) 16

RRB Group-D – 16/11/2018 (Shift-I)

Ans. (b) As per question,

$c - a = k$ ...(i)

$d - b = k$ ...(ii)

on subtracting

$c - a - d + b = 0$

$b + c = a + d$ ...(iii)

$\because$ mean = 25

$\frac{a+b+c+d}{4} = 25$

$a + (b + c) + d = 100$

$a + d + a + d = 100$ [from equation (iii)]

$a + d = 50$ ...(iv)

but $d - a = 20$

$d = 20 + a$

$\therefore$ $a + 20 + a = 50$

$2a = 30$

$a = 15$

130. 4 out of 5 cricketers have played the innings of 13, 9, 5, 11 respectively. If the mean of the data set is 9. Then what will be the number of innings played by that 5th player?

(a) 9 (b) 8

(c) 7 (d) 6

RRB NTPC 05.04.2016 Shift-1

Ans : (c) Total number of innings played by all five players= 5×9 = 45

Total number of innings of four Players

= 13+9+5+11 = 38

$\therefore$ Fifth player's total number of innings = 45–38 = 7

131. The mean of 8 observations is 10.5. In the given observation of the seven observations are 3, 15, 7, 19, 12, 17 and 8. Find the 8th observation-

(a) 10 (b) 11

(c) 3 (d) 12

RRB NTPC 04.04.2016 Shift : 1

Ans : (c) Sum of seven observation = 3 + 15 + 7 + 19 + 12 + 17 + 8 = 81

$\therefore$ eight observation= 8 × 10.5 – 81

= 84.0 – 81 = 3

132. What will be the mean, mode and median of the given numbers– 3, 4, 5, 3, 6, 3, 4, 5, 3

(a) 4, 4, 4 (b) 4, 4, 3

(c) 3, 4, 4 (d) 4, 3, 4

RRB NTPC 31.03.2016 Shift : 1

Ans : (d) Writing data in ascending order

3, 3, 3, 3, 4, 4, 5, 5, 6

$\text{Mean} = \frac{3+3+3+3+4+4+5+5+6}{9} = \frac{36}{9} = 4$

Mode = 3 (the highest frequent number)

$\because$ Terms number (n) = 9 (odd)

$\therefore \text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{term} = \left(\frac{9+1}{2}\right)^{th} \text{term}$

$= 5^{th}$ term = 4

133. What will be the median, mode and mean of the given numbers?

9, 5, 8, 9, 9, 7, 8, 9, 8

(a) 9, 9, 9 (b) 9, 8, 9

(c) 8, 9, 8 (d) 8, 9, 9

RRB NTPC 31.03.2016 Shift : 2

Ans : (c) Writing data in ascending order,

5, 7, 8, 8, 8, 9, 9, 9, 9

$\because$ terms number = 9 (odd)

$\therefore \text{median} = \left(\frac{n+1}{2}\right)$ th term

$= \frac{9+1}{2} = 5^{th}$ term = 8

mode = 9 (the highest frequent number)

$\text{mean} = \frac{5+7+8+8+8+9+9+9+9}{9}$

$= \frac{72}{9} = 8$

134. What will be the mode and median of the given digits–

8, 6, 8, 7, 8, 6, 8, 7, 6

(a) 7 and 8 (b) 6 and 7
(c) 8 and 7 (d) 6 and 8

RRB NTPC 30.03.2016 Shift : 1

Ans : (c) writing number in ascending order,
6, 6, 6, 7, 7, 8, 8, 8, 8
mode = 8 (most frequent)
terms number = 9 (odd)

$\therefore$ median $= \left(\frac{9+1}{2}\right) = 5^{th}$ term = 7

135. What will be range, mode and median of the given following data?

13, 14, 13, 12, 15, 21, 16, 18, 13

(a) 9, 13, 14 (b) 6, 13, 14
(c) 8, 13, 14 (d) 5, 13, 14

RRB Paramedical Exam – 20/07/2018 (Shift-III)

Ans : (a) Range = 21–12 = 9
Mode = 13 (Most frequent)
Writing data in ascending order
12, 13, 13, 13, 14, 15, 16, 18, 21

$\because$ n = 9 (odd)

$\therefore$ Median $= \left(\frac{9+1}{2}\right)^{th}$ term = 5th term = 14

136. What will be the mean and mode of the given following data?

1, 9, 5, 4, 2, 1, 9, 9, 2, 1, 9, 1, 2, 1

(a) 4 and 9 (b) 5 and 1
(c) 4 and 1 (d) 5 and 9

RRB NTPC 29.03.2016 Shift : 3

Ans : (c) Mean

$$= \frac{1+9+5+4+2+1+9+9+2+1+9+1+2+1}{14}$$

$$= \frac{56}{14} = 4$$

Mode of data = 1(term of most frequent)

137. The mean of a distribution is 13 and the standard deviation is 7. What is the value of variance coefficient?

(a) 50% (b) 76.77%
(c) 53.85% (d) 38.88%

RRB NTPC 19.04.2016 Shift : 1

Ans : (c)

Variance coeffcient $= \frac{\text{Standard deviation}}{\text{mean}} \times 100\%$

$$= \frac{7}{13} \times 100 = 53.85\%$$

138. The mean of 20 observations is 19. One another observation is included and the new mean is 20. What is 21st observation?

(a) 20 (b) 30
(c) 40 (d) 42

RRB NTPC 19.04.2016 Shift : 2

Ans : (c) Total sum of 20 observations=20 × 19 = 380
Sum on inclusion of new observation = 21 × 20 = 420
$\therefore$ Value of 21st observation = 420 – 380 = 40

139. The mean of a distribution is 18 and the standard deviation is 4.5. What is the value of variance coefficient?

(a) 50% (b) 25%
(c) 100% (d) 75%

RRB NTPC 19.04.2016 Shift : 2

Ans : (b) Distribution mean = 18
Standard deviation = 4.5

$\therefore$ Variance coefficient $= \frac{4.5}{18} \times 100\ \%$

$$= \frac{450}{18} = 25\%$$

140. The mean of a distribution is 11 and the standard deviation is 5. What is the value of variance coefficient?

(a) 45.45% (b) 35.35%
(c) 25.25% (d) 55.55%

RRB NTPC 18.04.2016 Shift : 1

Ans : (a) To calculate the variance coeffi divide the standard deviation by the mean and multiply by 100. or

Variance coefficient $= \frac{\text{Variance}}{\text{coefficient}}$

$$= \frac{\sigma}{x} \times 100\%$$

$$= \frac{5}{11} \times 100\%$$

$$= 45.45\%$$

141. The mean of a distribution is 10 and the standard deviation is 5. What is the value of variance coefficient?

(a) 50% (b) 100%
(c) 150% (d) 200%

RRB NTPC 18.04.2016 Shift : 2

Ans : (a)

Variance coefficient $= \frac{\text{Standard deviation}}{\text{mean}} \times 100\%$

$$= \frac{5}{10} \times 100 = 50\%$$

142. If the standard deviation of a population is 8, then what will be its variance?

(a) 64 (b) 16
(c) 32 (d) 24

RRB NTPC 18.04.2016 Shift : 2

Ans : (a) Standard deviation of variance coefficient= 8
Variance = (Standard deviation)^2
$= (8)^2 = 64$

143. The mean of a distribution is 24 and the standard deviation is 8. What is the value of variance coefficient?

(a) 16.66% (b) 66.66%
(c) 33.33% (d) 100%

RRB NTPC 18.04.2016 Shift : 3

Ans : (c) Mean of distributer = 24
Standard deviation= 8

$$\text{Variance coefficient} = \frac{\text{Standard deviation}}{\text{mean}} \times 100\%$$

$$= \frac{8}{24} \times 100\,\%$$

$$= \frac{100}{3}\,\%$$

$= 33.33\%$

144. The mean of a distribution is 20 and the standard deviation is 4, so find out its variance coefficient?

(a) 10 (b) 20
(c) 40 (d) 60

RRB NTPC 16.04.2016 Shift : 1

Ans : (b) Mean = 20
Standard deviation = 4

$$\therefore \text{Variance coefficient} = \frac{\text{Standard deviation}}{\text{mean}} \times 100\%$$

$$= \frac{4}{20} \times 100 = 20\%$$

145. The mean of a distribution is 18 and the standard deviation s 7. What is the value of variance coefficient?

(a) 50% (b) 76.77%
(c) 54.44% (d) 38.88%

RRB NTPC 16.04.2016 Shift : 2

Ans : (d) Mean = 18
Standard deviation = 7

$$\therefore \text{Variance coefficient} = \frac{\text{Standard deviation}}{\text{mean}} \times 100\%$$

$$= \frac{7}{18} \times 100\,\%$$

$= 700/18 = 38.88\%$

146. The mean of a distribution is 21 and the standard deviation is 7. Find the variance coefficient?

(a) 16.66% (b) 66.66%
(c) 33.33% (d) 100%

RRB NTPC 16.04.2016 Shift : 3

Ans : (c) Variance coefficient $= \frac{7}{21} \times 100 = 33.33\%$

147. The average of the results of 35 trials is 21. The average result of first 17 trials and last 17 trials are 19 and 22 respectively. What is the value of the result of the 18th test?

(a) 42 (b) 36
(c) 38 (d) 34

RRB NTPC 12.04.2016 Shift : 2

Ans : (c) average of 35 observations = 21
sum of 35 observations = 35×21 = 735
average of 17 observations = 19
sum total of 17 observations = 17×19 = 323
average of last 17 observations = 22
sum total of these = 17×22 = 374
Value of 18th obsevation = 735 – 374 – 323 = 38

148. The mean of a distribution is 15 and the standard deviation is 5. What is the value of variance coefficient?

(a) 16.66% (b) 66.66%
(c) 33.33% (d) 100%

RRB NTPC 12.04.2016 Shift : 3

Ans : (c)

$$\text{Variance coefficient} = \frac{\text{Standard deviation}}{\text{mean}} \times 100\%$$

$$= \frac{5}{15} \times 100$$

$$= \frac{100}{3} = 33.33\%$$

149. If the value of the mode is 14 and arithmetic mean is 5, then find the value of median?

(a) 8 (b) 18
(c) 12 (d) 14

RRB NTPC 10.04.2016 Shift : 3

Ans : (a) mode = 3 × median – 2 × mean
14 = 3× median – 2 × 5
3× median = 14 + 10

$$\text{median} = \frac{24}{3} = 8$$

150. What is the value of mode as well as median of the given following numbers–

3, 4, 5, 5, 3, 6, 7, 3, 5, 5, 6

(a) 5 and 5 (b) 3 and 5

(c) 5 and 4 (d) 3 and 4

RRB NTPC 09.04.2016 Shift : 3

Ans : (a) 3, 4, 5, 5, 3, 6, 7, 3, 5, 5, 6

Writing in ascending order = 3,3,3,4,5,5,5,5,6,6,7

mode = 5 (most frequent)

$\therefore$ where n = 11(odd)

Median = $\left(\frac{n+1}{2}\right) = \frac{11+1}{2} = 6^{th}$ term = 5

151. The variance of a given data is 324, then what will be the value of standard deviation?

(a) ±18 (b) 18

(c) 324 (d) 162

RRB NTPC 07.04.2016 Shift : 2

Ans : (b) Standard deviation $= \sqrt{\text{Variance}}$

$\therefore$ Variance = 324

$\therefore$ Standard deviation $= \sqrt{324}$

$= \pm 18$

$= 18$

152. If the mean value of 12 men's height is 1.70 m. and the mean value of 8 women of height is 1.60 m. What will be the sum of the total height of 8 women (in metre).

(a) 12.9 (b) 12.8

(c) 12.4 (d) 13

RRB NTPC 19.01.2017 Shift : 1

Ans : (b) Sum = number × mean

8 women's total height = 8×1.60 = 12.8m

153. The mean of a distribution is 80 and the standard deviation is 16. What is the value of variance coefficient?

(a) 10% (b) 20%

(c) 40% (d) 60%

RRB NTPC 22.04.2016 Shift : 1

Ans : (b) distribution mean = 80

Standard deviation = 16

Variance coefficient $= \frac{\text{Standard deviation}}{\text{distributers}} \times 100\%$

$= \left(\frac{16}{80}\right) \times 100\ \%$

$= \frac{1}{5} \times 100\ \%$

$= 20\%$

154. If the standard deviation of the population is 13, then what will be the variance of the population?

(a) 78 (b) 39

(c) 26 (d) 169

RRB NTPC 22.04.2016 Shift : 3

Ans : (d) Required variance = $(\text{standard deviation})^2$

$= 13 \times 13 = 169$

155. What is the value of median, mode and mean of the given following numbers?

9, 8, 3, 5, 1, 9, 8, 2, 9

(a) 9, 9, 6 (b) 9, 6, 9

(c) 8, 9, 6 (d) 8, 5, 6

RRB NTPC 26.04.2016 Shift : 1

Ans : (c) ascending order

1, 2, 3, 5, 8, 8, 9, 9, 9

n = 9 (odd)

Median $= \left(\frac{n+1}{2}\right)$ th term $= \left(\frac{9+1}{2}\right)^{th}$ term

$= 5^{th}$ term = 8

Mean $= \frac{\text{sum of all numbers}}{\text{Total numbers}}$

$= \frac{9+8+3+5+1+9+8+2+9}{9} = \frac{54}{9} = 6$

Mode = 9 (Most frequent)

Hence median mode and mean is 8, 9, 6.

156. If the standard deviation of a distribution is 9. Then find the value of variance.

(a) 18 (b) 27

(c) 81 (d) 36

RRB NTPC 28.04.2016 Shift : 2

Ans : (c) Variance = $(\text{Standard deviation})^2$

$= (9)^2 = 81$

157. The mean length of 6 rods is 44.2 cm. If the mean length of 5 rods is 46 cm. then what is the value of length of the 6th rods?

(a) 35 (b) 35.2

(c) 35.1 (d) 35.5

RRB NTPC 30.04.2016 Shift : 3

Ans : (b) Mean length of 6 rods = 44.2 cm.

Total length of 6 rods = 44.2 × 6 = 265.2

Mean length of 5 rods = 46 cm

Total length of 5 rods = 46 × 5 = 230

Length of 6th rod = 265.2 – 230 = 35.2 cm